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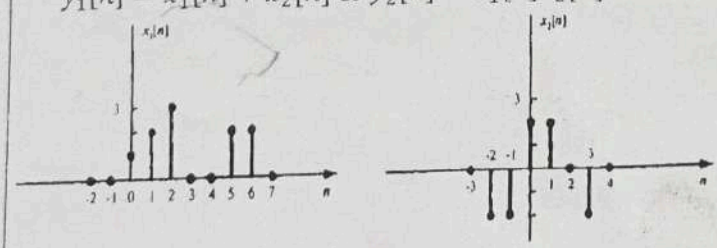
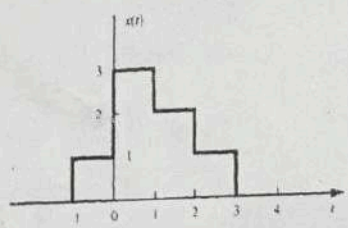
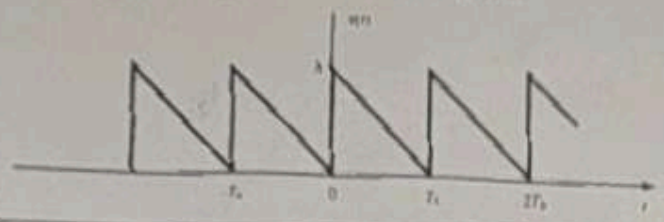
राष्ट्रीय प्रौद्योगिकी संस्थान, कुरुक्षेत्र
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
DEPARTMENT OF ELECTRICAL ENGINEERING
THEORY EXAMINATION
Question Paper

Month and year of the Examination	:	Dec-2025	Roll No.	:	
Programme	:	B.Tech	Semester	:	3rd
Subject	:	Signals and Systems			
Course No	:	RAPC204	Maximum Marks	:	50
Number of Questions to be attempted	:	5	Time allowed	:	3 hr
Total No. of Questions	:	6	Total No. of Pages used	:	2

Unless stated otherwise, the symbols have their usual meanings in context with the subject.
Assume suitably and state, additional data required, if any.
The candidate, before starting to write the solutions, should please check the Question Paper for any discrepancy, and ensure that they have been delivered the question paper of right course no. and right subject title.

Note: First Question is compulsory

1.	a) Check whether the signal $x(t) = tu(t)$ is an energy or power signal? b) Check for the periodicity of a signal $x(t) = \cos\frac{\pi}{3}t + \sin\frac{\pi}{4}t$ and obtain the resultant period if the signal is a periodic signal? c) Compute $t\delta(t)$, $\sin t\,\delta(t)$, $\cos t\,\delta(t-\pi)$ and $\int_{-\infty}^{\infty}(t^2+\cos\pi t)\delta(t-1)dt$? d) Determine the even and odd components of the signal $x(t) = \sin(\omega_0 t + \frac{\pi}{4})$ e) Consider the system with input-output relation ship $y[n] = \frac{1}{2}(x[n] + x[n-1])$. Compute the response of a system to the input of $x[n] = 2\delta[n-1] + 2\delta[n-2]$?	[5*2 M]
2.	a) Consider the discrete time signals $x_1[n]$ and $x_2[n]$ shown in the Fig-1. Compute $y_1[n] = x_1[n] + x_2[n]$ & $y_2[n] = x_1[n]x_2[n]$ and draw the resultant graphs?  Fig-1  Fig-2 b) Express the signal shown in the Fig-2 in terms of unit step functions along with all graphical representations?	[5+5 M]
3.	a) Find the inverse Laplace transform of $X(s) = \frac{2+2se^{-2s}+4e^{-4s}}{s^2+4s+3}$ $Re\{s\} > -1$? b) The output $y(t)$ of a continuous-time LTI system is found to be $2e^{-3t}u(t)$ when the input $x(t) = u(t)$. Find the impulse response $h(t)$ of the system and output $y(t)$ when the input $x(t) = e^{-t}u(t)$?	[5+5 M]

4.	a) Determine the convolution $y(t) = x(t) \cdot h(t)$ of the signals $x(t) = u(t) - u(t-3)$ and $h(t) = u(t) - u(t-2)$ by graphical method and draw the final output? b) Obtain the Fourier series of the signal shown in the figure? 	[6+4 M]
5.	a) State sampling theorem and Prove that how will you recover the original signal after the sampling? b) Prove that Fourier Transform maps the convolution of two signals into the product of their Fourier Transforms. Use this property to obtain the input $x(t)$ for causal LTI system output $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$ and frequency response $H(j\omega) = \frac{1}{jw+3}$?	[5+5 M]
6.	a) Proof the Z-transform and Fourier series properties of **convolution in time** and **multiplication in time**? b) Find the Z-Transform of the sequence $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[-n-1]$ and sketch the pole-zero plot with ROC?	[6+4 M]

**** All the Best ****